Online 3 Hour Test

Q. No		Marks
	Part A	
1	Answer the following:	
1.1	For Maximize $Z = 3x_1 + 4x_2$ subject to the constraints: $x_1 + x_2 \leq 4$, $x_1 \geq 0$, $x_2 \geq 0$, what will be the value of Z.	(01)
1.2	Decision variables in optimization problems represent the _____ factors or parameters whose values are to be determined in order to optimize a certain objective.	(01)
1.3	The simplex method efficiently identifies the optimal solution by traversing through the _____ of the feasible region	(01)
1.4	The primary objective of the two-phase simplex method is to find a feasible solution before transitioning to the _____ phase.	(01)
1.5	Sensitivity analysis in linear programming examines the impact of changes in _____ coefficients, constraints, or other parameters on the optimal solution	(01)
1.6	Sensitivity analysis in linear programming assesses the impact of changes in _____ on the optimal solution and associated variables.	(01)
1.7	In linear programming, the concept of duality establishes a relationship between a primal problem and its corresponding _____ problem.	(01)
1.8	In the dual simplex method, each iteration involves pivoting on a non-basic variable to improve feasibility and bring the _____ variable into the basis.	(01)
1.9	In the transportation problem, the objective is to minimize the total _____ of transporting goods from suppliers to destinations	(01)
1.10	The transportation problem can be formulated as a _____ programming model.	(01)
1.11	The assignment problem can be solved using _____ algorithms such as the Hungarian algorithm.	(01)
1.12	The Travelling Salesman Problem can be formulated as a _____ optimization problem	(01)
1.13	In graphical sensitivity analysis, the _____ is the line representing the objective function value.	(01)
1.14	In CPM (Critical Path Method), the critical path represents the longest sequence of activities that determines the minimum _____ of the project.	(01)
1.15	PERT (Program Evaluation and Review Technique) is a project management tool used to estimate the _____ time for completing each task	(01)
1.16	In the context of project management, floats represent the _____ time available for an activity without delaying the project's completion.	(01)
1.17	In a zero-sum game, the total gains of the participants equal the total _____.	(01)
1.18	In game theory, the dominance rule states that if one strategy of a player is always better than another strategy, regardless of the other player's actions, then the dominated strategy can be _____ from consideration.	(01)
1.19	Game theory is a branch of mathematics and economics that studies _____ interactions where the outcome of one participant's decision depends on the decisions of others.	(01)
1.20	The graphical method in game theory is a visual tool used to analyze and illustrate the possible _____ of strategies for each player in a game.	(01)
	Part B	
2 a.	Explain the seven step model building process with an example for each step.	(06)

b.	An organization has Rs. 40 lakhs to contribute and is to pick among three kinds of vehicles A, B and C. Vehicle A has 10 – ton payload and is required to average 30 km each hour. It costs Rs. 70000. Vehicle B has a 22 – ton payload, is required to average 40 km each hour and costs Rs. 90000. Vehicle C is an altered type of B. It has resting space for the driver, which lessens its payload ability to 15 – ton, while raising the expense for Rs. 100000. Vehicle A requires a group of one man, and whenever driven on three movements each day, could run for a normal of 22 hours per day. Vehicle B and C require a team of two men each and whenever driven on three movements each day, could be run for a normal 21 hours and 18 hours individually. The organization has an armada of 80 crew members accessible to it. In the event that the absolute number of vehicles are not to surpass 60, the number of vehicles of each sort ought to be bought if the organization needs to boost its ability in ton km each day? Formulate the problem as L. P. problem.	(10)
3 a.	List the assumptions of linear programming problem.	(04)
b.	Solve the following linear programming problem by simplex method: Maximize $Z = 5x_1 + 4x_2$ Subject to the constraints: $6x_1 + 4x_2 \leq 24$ $6x_1 + 2x_2 \leq 6$ $-x_1 + x_2 \leq 1$ $x_1, x_2 \geq 0$	(12)
	OR	
4 a.	Discuss the characteristics of duality.	(06)
b.	Use dual simplex method to solve the following linear programming problem: Minimize $Z = 2x_1 + x_2$ Subject to $3x_1 + x_2 \geq 3$ $4x_1 + 3x_2 \geq 6$ $x_1 + 2x_2 \geq 3$ and $x_1 \geq 0, x_2 \geq 0$	(12)
5 a.	Find the initial basic feasible solution for the following transportation problem, using North – West Corner Rule method. Destination / Supply D1 D2 D3 S1: 3 8 5 — 7 S2: 4 4 2 — 8 S3: 6 5 8 — 10 S4: 2 6 3 — 15 Sources Demand: 8 10 22	(04)
b.	A product is produced by four factories A, B, C and D. The unit production costs in them are Rs. 2, Rs. 3, Re. 1 and Rs. 5 respectively. Their production capacities are factory A – 50 units, B – 70 units, C – 30 units and D – 50 units. These factories supply the product to four stores, demands of which are 25, 35, 105 and 20 units respectively. Unit transport cost in rupees from each factory to each store is given in the table below: Stores 1 2 3 4 A: 2 4 6 11 B: 10 8 7 5 C: 13 3 9 12 D: 4 6 8 3 Factories Determine the extent of deliveries from each of the factories to each of the stores so that the total production and transportation cost is minimum.	(12)

OR

6 a. Solve the following assignment problem shown in Table using Hungarian method. The matrix entries are processing time of each Job to each machine in hours. (08)

J/M	I	II	III	IV	V
1	9	22	58	11	19
2	43	78	72	50	63
3	41	28	91	37	45
4	74	42	27	49	39
5	36	11	57	22	25

b. A travelling salesman has to visit five cities. He wishes to start from a particular city, visit each city once and then return to his starting point. The travelling cost of each city from a particular city is given below:

	A	B	C	D	E
A	∞	2	5	7	1
B	6	∞	3	8	2
C	8	7	∞	4	7
D	12	4	6	∞	5
E	1	3	2	8	∞

What is the sequence of visit of the salesman, so that the cost is minimum? (08)

7 a. Distinguish between CPM and PERT of network analysis based on orientation, evolution, model and estimates. (08)

b. Draw the network diagram, identify critical path and total project duration for the following: (08)

Activity	1 – 2	1 – 3	1 – 4	2 – 5	3 – 7	4 – 6	5 – 7
Duration	10	8	9	8	16	7	7
Activity	5 – 8	6 – 7	6 – 9	7 – 10	8 – 10	9 – 10	
Duration	6	7	5	12	13	15	

OR

8 a. Explain the difference between Activity-on-Node (AON) and Activity-on-Arrow (AOA) representations in network diagrams. (06)

b. The following network diagram represents activities associated with a project: (12)

Activity	A	B	C	D	E	F	G	H	I
Optimistic Time	5	18	26	16	15	6	7	7	3
Pessimistic Time	10	22	40	20	25	12	12	9	5
Most Likely Time	8	20	33	18	20	9	10	8	4

i. Find expected completion time and variance of each activity
ii. Draw Network diagram
iii. The critical path and project duration
iv. The duration of the project that will have 95 per cent chance of being completed.

9 a. List the assumptions of game. (06)

b. Use graphical method in solving the following game and find the value of the game. (10)

		Player B	
		B_1	B_2
Player A	A_1	-2	4
	A_2	8	3
	A_3	9	0

OR

10 a. Define the following with respect to game theory: (08)
i. Two person zero sum game
ii. Pay-off Matrix
iii. Saddle point

b.	A company is currently involved in negotiations with its union on the upcoming wage contract. Positive signs in table represent wage increase while negative sign represents wage reduction. What are the optimal strategies for the company as well as the union? What is the game value? Solve by dominance property.(Rs. In lakhs)	(08)

Conditional cost to the company

		Union Strategies			
		U_1	U_2	U_3	U_4
Company Strategies	C_1	0.25	0.27	0.35	- 0.02
	C_2	0.20	0.06	0.08	0.08
	C_3	0.14	0.12	0.05	0.03
	C_4	0.30	0.14	0.19	0.00